We will begin around 8:05am. There will be important announcement about the tutorial and lecture hour at the beginning, please do not go I need your full attention and response on this.

Lecture 3: Hyperbolic Functions

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

Sem II, 2025/2026

Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Understand what is the hyperbolic function.
- ☐ Find the derivative and the integral of the hyperbolic function.
- ☐ Find the inverse hyperbolic function.
- ☐ Find the derivative of the inverse hyperbolic function.
- ☐ Perform basic integration using integration formula involving inverse hyperbolic function.

Introduction

Any function $f : \mathbb{R} \to \mathbb{R}$ can be written as a sum of an even function and an odd function.

$$f(x) = \frac{f(x) + f(-x)}{2} + \frac{f(x) - f(-x)}{2}$$

Let $g(x) = \frac{f(x)+f(-x)}{2}$ even

$g(-x) = \frac{f(-x)+f(x)}{2} = g(x) \to$ even.

Let $h(x) = \frac{f(x)-f(-x)}{2}$ odd.

$h(-x) = \frac{f(-x)-f(x)}{2} = -h(x)$ → odd.

If we let $f(x) = e^x$, then we have

$$e^x = \frac{e^x + e^{-x}}{2} + \frac{e^x - e^{-x}}{2}$$

$\cosh x$ (even)

$\sinh x$ (odd)

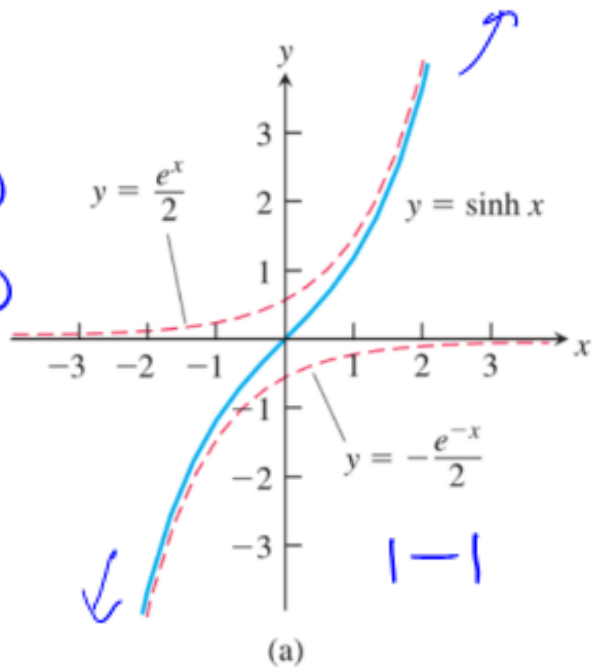
Hyperbolic Functions

Definition: We define the hyperbolic sine and hyperbolic cosine functions by the odd and even parts of $f(x) = e^x$, respectively.

$$\sinh x = \frac{e^x - e^{-x}}{2} \quad \text{and} \quad \cosh x = \frac{e^x + e^{-x}}{2}.$$

D: $(-\infty, \infty)$

R: $(-\infty, \infty)$



(a)

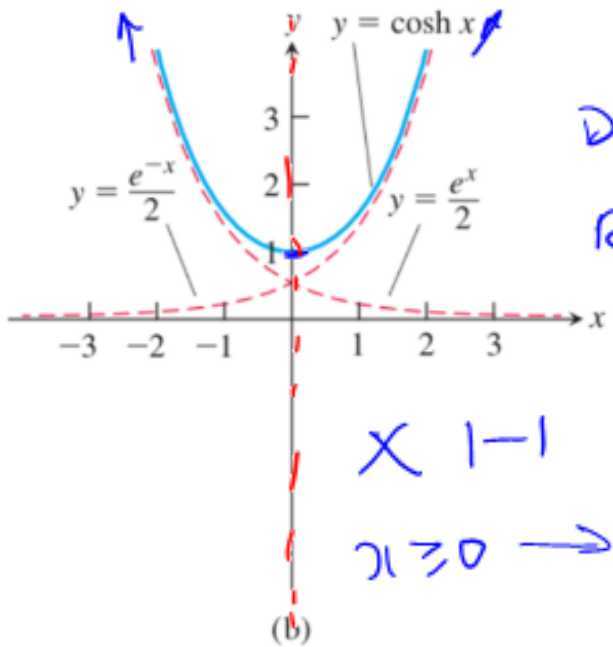
1−1

Hyperbolic sine:

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

D: $(-\infty, \infty)$

R: $[1, \infty)$



(b)

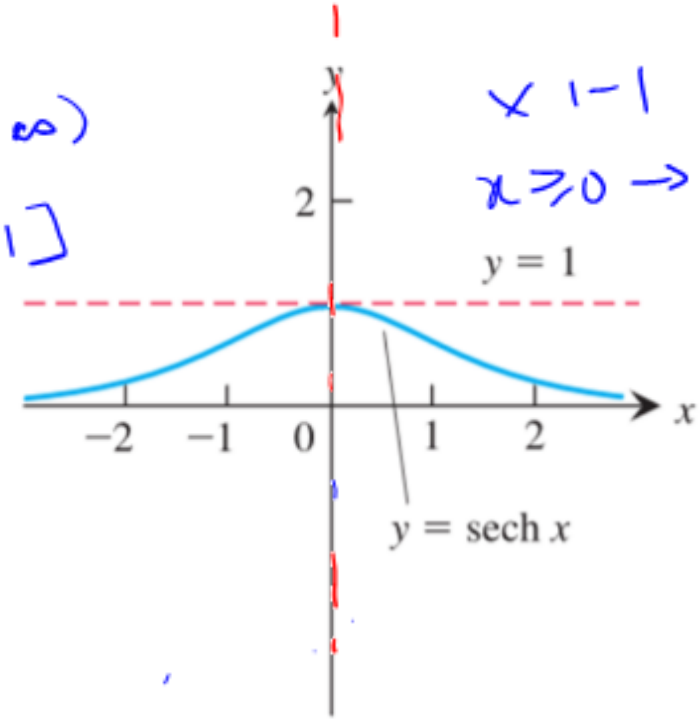
X 1−1

$x \geq 0 \rightarrow$ 1−1

Hyperbolic cosine:

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

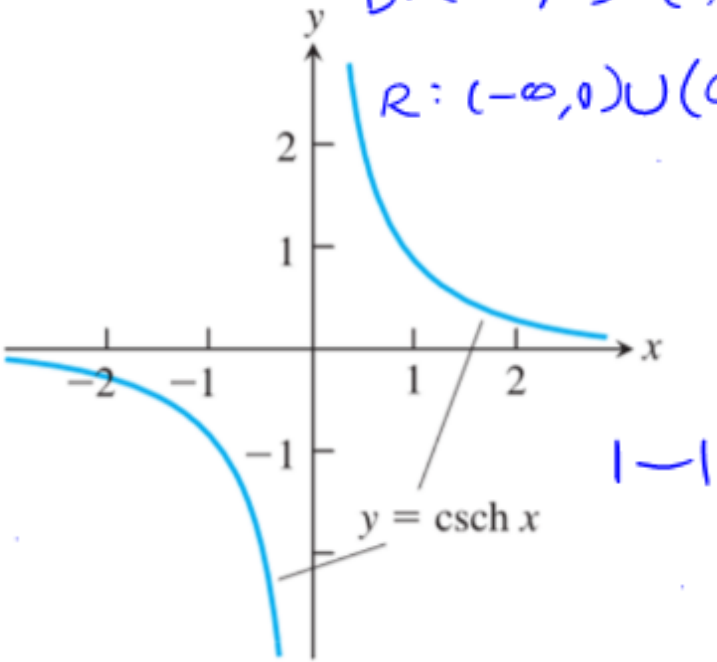
Hyperbolic Functions



(d)

Hyperbolic secant:

$$\operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}$$

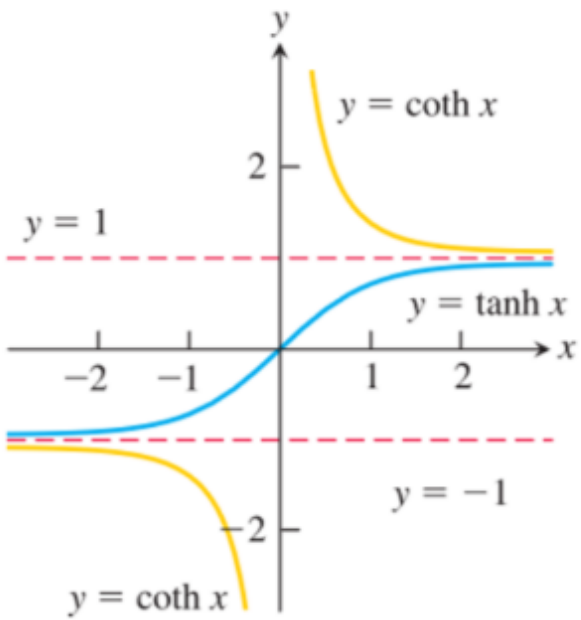


(e)

Hyperbolic cosecant:

$$\operatorname{csch} x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}}$$

Hyperbolic Functions



(c)

Hyperbolic tangent:

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

D: $(-\infty, \infty)$
R: $(-1, 1)$
1-1

Hyperbolic cotangent:

$$\coth x = \frac{\cosh x}{\sinh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}$$

D: $(-\infty, 0) \cup (0, \infty)$
R: $(-\infty, -1) \cup (1, \infty)$
1-1

$\frac{1}{\tanh x}$

Hyperbolic Functions

Some identities for hyperbolic functions

- $\cosh^2 x - \sinh^2 x = 1$

$$\cosh^2 x - \sinh^2 x = \left(\frac{e^x + e^{-x}}{2}\right)^2 - \left(\frac{e^x - e^{-x}}{2}\right)^2$$

$$= \frac{(e^x + e^{-x})^2}{4} - \frac{(e^x - e^{-x})^2}{4}$$

$$= \frac{e^{2x} + 2 + e^{-2x}}{4} - \frac{e^{2x} - 2 + e^{-2x}}{4}$$

$$= \frac{e^{2x} + 2 + e^{-2x} - e^{2x} + 2 - e^{-2x}}{4}$$

$$= 4/4 = 1$$

- $\sinh 2x = 2 \sinh x \cosh x$

Exercise.

- $\cosh 2x = \cosh^2 x + \sinh^2 x$

$$\cosh^2 x + \sinh^2 x = \left(\frac{e^x + e^{-x}}{2}\right)^2 + \left(\frac{e^x - e^{-x}}{2}\right)^2$$

$$= \frac{e^{2x} + e^{-2x}}{2} = \cosh 2x$$

- $\cosh^2 x = \frac{\cosh 2x + 1}{2}$

$$\cosh 2x = \cosh^2 x + \sinh^2 x$$

$$= \cosh^2 x + (\cosh^2 x - 1) = 2\cosh^2 x - 1$$

$$\Rightarrow \cosh^2 x = \frac{\cosh 2x + 1}{2}$$

Hyperbolic Functions

Some identities for hyperbolic functions

- $\sinh^2 x = \frac{\cosh 2x - 1}{2}$
- $\tanh^2 x = 1 - \operatorname{sech}^2 x$
- $\coth^2 x = 1 + \operatorname{csch}^2 x$

Exercise:
Try to verify these.
(refer to previous identities)

Derivatives of Hyperbolic Functions

1. $\frac{d}{dx}(\sinh x) = \cosh x$
2. $\frac{d}{dx}(\cosh x) = \sinh x$ — Exercise.
3. $\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$
4. $\frac{d}{dx}(\coth x) = -\operatorname{csch}^2 x$
5. $\frac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \tanh x$
6. $\frac{d}{dx}(\operatorname{csch} x) = -\operatorname{csch} x \coth x$

$$\frac{d}{dx}(\sinh x) = \frac{d}{dx}\left(\frac{e^x - e^{-x}}{2}\right)$$

$$= \frac{1}{2}\left[e^x - (-e^{-x})\right]$$

$$= \frac{e^x + e^{-x}}{2} = \cosh x.$$

$$\frac{d}{dx}(\tanh x) = \frac{d}{dx}\left(\frac{\sinh x}{\cosh x}\right)$$

$$= \frac{\frac{d}{dx}(\sinh x)\cosh x - \frac{d}{dx}(\cosh x)\sinh x}{\cosh^2 x}$$

Exercise.
Hint: Use DEF + quotient rule.

$$= \frac{\cosh^2 x - \sinh^2 x}{\cosh^2 x}$$

$$= \frac{1}{\cosh^2 x} = \operatorname{sech}^2 x$$

Derivatives of Hyperbolic Functions

Example: Evaluate

1. $\frac{d}{dx}\cosh(1+x^2)$ $= \sinh(1+x^2)\frac{d}{dx}(1+x^2) = 2x\sinh(1+x^2)$

2. $\frac{d}{dx}(\tanh\sqrt{1+x^2})$ $= \operatorname{sech}^2\sqrt{1+x^2}\ \frac{d}{dx}\left(\sqrt{1+x^2}\right) = \left(\operatorname{sech}^2\sqrt{1+x^2}\right)\left(\frac{1}{2\sqrt{1+x^2}}\right)\frac{d}{dx}(1+x^2)$ ($2x$)

$= \dfrac{x\operatorname{sech}^2\sqrt{1+x^2}}{\sqrt{1+x^2}}$

3. $\frac{d}{dx}(x^2\sinh x) = 2x\sinh x + x^2\cosh x$ ($\frac{d}{dx}x^2$, $\frac{d}{dx}\sinh x$)

4. $\frac{d}{dx}(e^x\sinh x) = e^x\sinh x + e^x\cosh x = e^x(\sinh x + \cosh x)$

$= e^x\left(\dfrac{e^x-e^{-x}}{2} + \dfrac{e^x+e^{-x}}{2}\right)$

$= e^x(e^x) = e^{2x}$

5. $\frac{d}{dx}[(x^2-1)\operatorname{csch}(\ln x)]$

Derivatives of Hyperbolic Functions

(5) $\frac{d}{dx}\left[(x^2-1)\operatorname{csch}(\ln x)\right]$

hyperbolic function. A $\ln x$ can be simplified using definition.

$$= \frac{d}{dx}\left[(x^2-1)\left(\frac{1}{\sinh(\ln x)}\right)\right]$$

$$= \frac{d}{dx}\left[(x^2-1)\left(\frac{2}{e^{\ln x} - e^{-\ln x}}\right)\right]$$

$\ln x^{-1} = \ln\left(\frac{1}{x}\right)$

$$= \frac{d}{dx}\left[(x^2-1)\left(\frac{2}{x-\frac{1}{x}}\right)\right] = \frac{d}{dx}\left[(x^2-1)\left[\frac{2}{\frac{1}{x}(x^2-1)}\right]\right]$$

$$= \frac{d}{dx}(2x) = 2.$$

Integrals of Hyperbolic Functions

1. $\int \sinh x \, dx = \cosh x + C$
2. $\int \cosh x \, dx = \sinh x + C$
3. $\int \operatorname{sech}^2 x \, dx = \tanh x + C$
4. $\int \operatorname{csch}^2 x \, dx = -\coth x + C$
5. $\int \operatorname{sech} x \tanh x \, dx = -\operatorname{sech} x + C$
6. $\int \operatorname{csch} x \coth x \, dx = -\operatorname{csch} x + C$

Integrals of Hyperbolic Functions

Example: Evaluate

1. $\int \coth 5x \, dx$

Let $u = \sinh 5x$

$du = 5\cosh 5x \, dx$

$= \int \frac{\cosh 5x}{\sinh 5x} \, dx = \frac{1}{5} \int \frac{5\cosh 5x \, dx}{\sinh 5x} = \frac{1}{5} \int \frac{du}{u}$

$= \frac{1}{5} \ln|u| + C$

$= \frac{1}{5} \ln|\sinh 5x| + C$

2. $\int_0^1 \sinh^2 x \, dx$

$= \int_0^1 \frac{\cosh 2x - 1}{2} \, dx$

$= \frac{1}{2} \int_0^1 \cosh 2x - 1 \, dx$

$= \frac{1}{2} \left[\frac{\sinh 2x}{2} - x \right]_0^1$

$= \frac{1}{2} \left[\frac{\sinh 2}{2} - 1 - 0 \right] \approx ?$ calculator.

3. $\int_0^{\ln 2} 4e^x \sinh x \, dx$

4. $\int_0^{\frac{\ln 2}{3}} \cosh 3x \, dx$

③ $\int_0^{\ln 2} 4e^x \sinh x \, dx = \int_0^{\ln 2} 4e^x \left(\frac{e^x - e^{-x}}{2}\right) dx$

$= \int_0^{\ln 2} 2e^{2x} - 2 \, dx$

$= \left[\frac{2e^{2x}}{2} - 2x\right]_0^{\ln 2}$

$= \left[e^{2(\ln 2)} - 2\ln 2\right] - \left[e^{2(0)} - 2(0)\right]$

$= 4 - 2\ln 2 - 1$

$= 3 - 2\ln 2.$

(4) $\int_0^{\frac{\ln 2}{3}} \cosh 3x \, dx$

Let $u = 3x$

$du = 3dx$

$x = 0 \Rightarrow u = 0$

$x = \frac{\ln 2}{3} \Rightarrow u = \ln 2$

$$= \int_0^{\ln 2} \frac{1}{3} \cosh u \, du$$

$$= \left[\frac{\sinh u}{3} \right]_0^{\ln 2}$$

$$= \frac{\sinh(\ln 2)}{3} - 0$$

$$= \frac{1}{3} \left[\frac{e^{\ln 2} - e^{-\ln 2}}{2} \right] = \frac{1}{6} \left[2 - \frac{1}{2} \right]$$

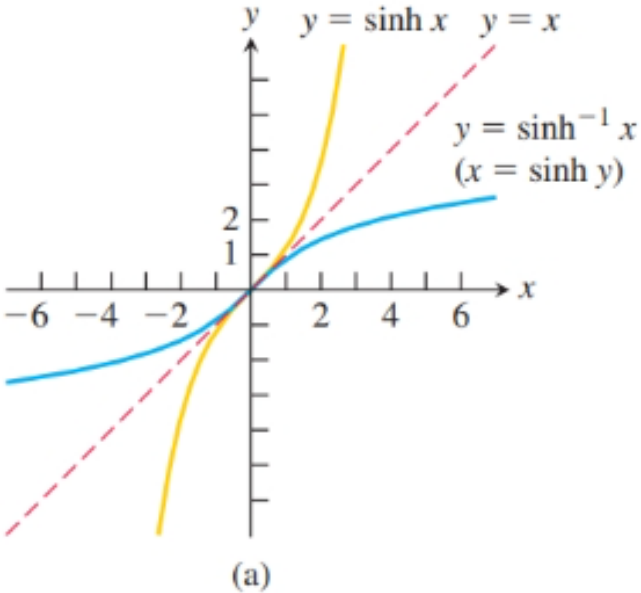
$$= 1/4.$$

Inverse Hyperbolic Function

*Do reflection on the line $y = x$

$\cosh x$ and $\operatorname{sech} x$ $\times$ 1-1

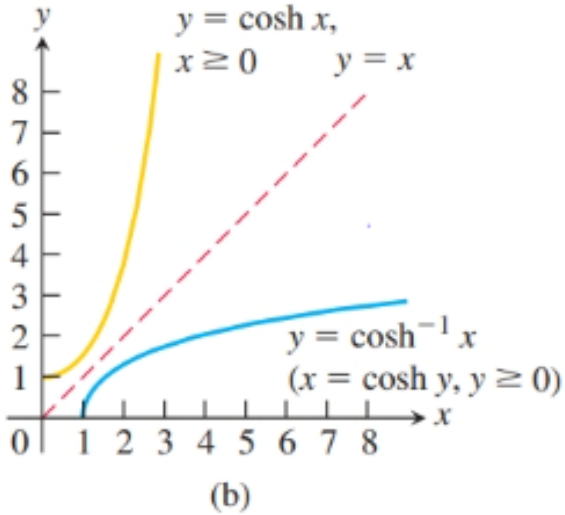
we restrict the domain to $[0, \infty)$



(a)

$D: (-\infty, \infty)$

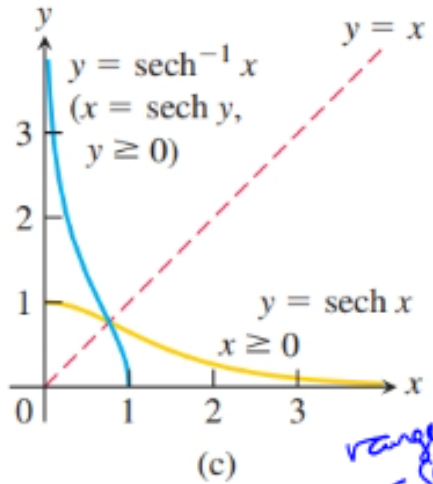
$R: (-\infty, \infty)$



(b)

$D: [1, \infty)$ ← range of $\cosh x$

$R: [0, \infty)$ ← restricted domain of $\cosh x$



(c)

range of $\operatorname{sech} x$

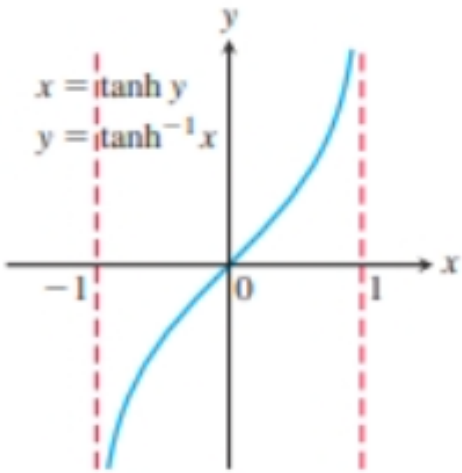
$D: (0, 1]$

$R: [0, \infty)$

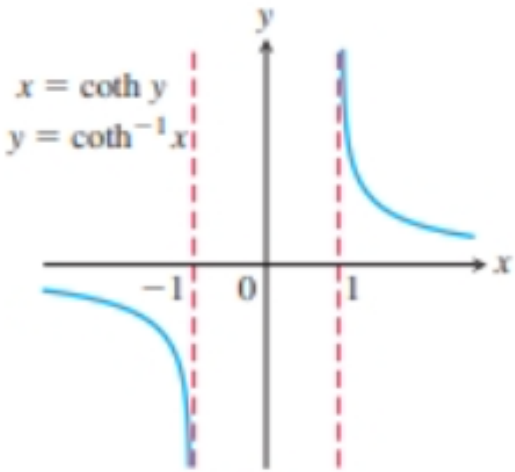
Inverse Hyperbolic Function

D: $(-\infty, 0) \cup (0, \infty)$

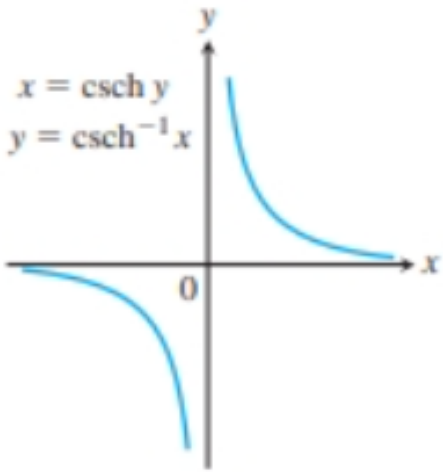
R: $(-\infty, 0) \cup (0, \infty)$



(a)



(b)



(c)

D: $(-1, 1)$

R: $(-\infty, \infty)$

D: $(-\infty, -1) \cup (1, \infty)$

R: $(-\infty, 0) \cup (0, \infty)$

Inverse Hyperbolic Functions

We know that $f: A \to B$, $f^{-1}: B \to A$.

$f(f^{-1}(x)) = x, \forall x \in B$. and $f^{-1}f(x) = x, \forall x \in A$.

$$\sinh(\sinh^{-1} x) = x \ \forall x \in \mathbb{R} \quad , \quad \sinh^{-1}(\sinh x) = x \ \forall x \in \mathbb{R}$$

$$\cosh(\cosh^{-1} x) = x \ \forall x \in [1, \infty) \quad , \quad \cosh^{-1}(\cosh x) = x \ \forall x \in [0, \infty)$$

($[1,\infty)$: range of $\cosh x$; $[0,\infty)$: range of $\cosh^{-1} x$)

$$\tanh(\tanh^{-1} x) = x \ \forall x \in [-1, 1] \quad , \quad \tanh^{-1}(\tanh x) = x \ \forall x \in \mathbb{R}$$

$$\operatorname{sech}(\operatorname{sech}^{-1} x) = x \ \forall x \in (0, 1] \quad , \quad \operatorname{sech}^{-1}(\operatorname{sech} x) = x \ \forall x \in [0, \infty)$$

$$\coth(\coth^{-1} x) = x \ \forall x \in \mathbb{R} \setminus [-1, 1] \quad , \quad \coth^{-1}(\coth x) = x \ \forall x \in \mathbb{R} \setminus \{0\}$$

$$\operatorname{csch}(\operatorname{csch}^{-1} x) = x \ \forall x \in \mathbb{R} \setminus \{0\} \quad , \quad \operatorname{csch}^{-1}(\operatorname{csch} x) = x \ \forall x \in \mathbb{R} \setminus \{0\}$$

Inverse Hyperbolic Functions

Some identities for inverse hyperbolic functions

- $\operatorname{sech}^{-1} x = \cosh^{-1}(\frac{1}{x}), x \in (0, 1]$

- $\operatorname{csch}^{-1} x = \sinh^{-1}(\frac{1}{x}), x \in (-\infty, 0) \cup (0, \infty)$

- $\coth^{-1} x = \tanh^{-1}(\frac{1}{x}), x \in (-\infty, -1) \cup (1, \infty)$

$$\sinh(x) = \frac{e^x - e^{-x}}{2}$$

Let $y = \sinh^{-1} x$.

$$\sinh(\sinh^{-1} x) = x$$

$$\sinh y = x$$

$$\frac{e^y - e^{-y}}{2} = x$$

$$e^y - e^{-y} = 2x$$

$\times e^y$, $e^{2y} - 1 = 2xe^y$

$$e^{2y} - 2xe^y - 1 = 0$$

Let $t = e^y$

$$t^2 - 2xt - 1 = 0$$

$$e^y = t = \frac{2x \pm \sqrt{4x^2+4}}{2}$$

$$= x \pm \sqrt{x^2+1}$$

$$= x + \sqrt{x^2+1} \quad (e^y > 0)$$

$$\sinh^{-1} x = y = \ln\left(x + \sqrt{x^2+1}\right)$$

Exercise :

Show (i) $\cosh^{-1}(x) = \ln\left(x + \sqrt{x^2 - 1}\right)$, $x \in [1, \infty)$

(ii) $\tanh^{-1}(x) = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$, $x \in (-1, 1)$

Derivatives of Inverse Hyperbolic Functions

1. $\frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{1+x^2}}$
2. $\frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2-1}}$ $(x > 1)$
3. $\frac{d}{dx}(\tanh^{-1} x) = \frac{1}{1-x^2}$ $(|x| < 1)$
4. $\frac{d}{dx}(\coth^{-1} x) = \frac{1}{1-x^2}$ $(|x| > 1)$
5. $\frac{d}{dx}(\operatorname{sech}^{-1} x) = -\frac{1}{x\sqrt{1-x^2}}$ $(0 < x < 1)$
6. $\frac{d}{dx}(\operatorname{csch}^{-1} x) = -\frac{1}{|x|\sqrt{1+x^2}}$ $(x \neq 0)$

$$\frac{d}{dx}(\sinh^{-1} x) = \frac{d}{dx}\left(\ln\left(x+\sqrt{x^2+1}\right)\right)$$

$$= \frac{1}{x+\sqrt{x^2+1}} \frac{d}{dx}\left(x+\sqrt{x^2+1}\right)$$

$$= \frac{1}{x+\sqrt{x^2+1}} \left(1 + \frac{1}{2\sqrt{x^2+1}}(2x)\right)$$

$$= \frac{1}{x+\sqrt{x^2+1}} \left[\frac{\sqrt{x^2+1}+x}{\sqrt{x^2+1}}\right]$$

$$= \frac{1}{\sqrt{1+x^2}}$$

Derivative of Inverse Hyperbolic Function

Example: Evaluate

1. $\frac{d}{dx}(\sinh^{-1} x^2)$ $= \frac{1}{\sqrt{1+(x^2)^2}} \frac{d}{dx}(x^2) = \frac{2x}{\sqrt{1+x^4}}$
2. $\frac{d}{dx}(\tanh^{-1}(\sin 2x))$ $= \frac{1}{1-(\sin 2x)^2} \frac{d}{dx}(\sin 2x) = \frac{1}{1-\sin^2(2x)}(2\cos 2x)$
3. $\frac{d}{dx}(\cosh^{-1}(x^2-1))$ $= \frac{2\cos 2x}{\cos^2(2x)}$
4. $\frac{d}{dx}(\coth^{-1}(3x^2))$ $= 2\sec(2x)$
5. $\frac{d}{dx}(\ln(x)\cosh^{-1}(x))$

③ $\frac{d}{dx}\left(\cosh^{-1}(x^2-1)\right) = \frac{1}{\sqrt{(x^2-1)^2-1}} \frac{d}{dx}(x^2-1)$, $x^2-1>1$, $x^2>2$

$$= \frac{2x}{\sqrt{x^4-2x^2}} = \frac{2x}{\sqrt{x^2(x^2-2)}}$$

$$= \frac{2x}{x\sqrt{x^2-2}} = \frac{2}{\sqrt{x^2-2}}$$

④ $\frac{d}{dx}\left(\coth^{-1}(3x^2)\right) = \frac{1}{1-(3x^2)^2} \frac{d}{dx}(3x^2)$

$$= \frac{6x}{1-9x^4}$$

⑤ $\frac{d}{dx}\left(\ln x \cosh^{-1} x\right) = \frac{1}{x}\cosh^{-1} x + \ln x \left(\frac{1}{\sqrt{x^2-1}}\right)$

Integration Formulas

Integrals evaluated with inverse ~~trigonometric~~ hyperbolic functions.

1. $\int \frac{dx}{\sqrt{a^2+x^2}} = \sinh^{-1}\left(\frac{x}{a}\right) + C, \qquad a > 0$

2. $\int \frac{dx}{\sqrt{x^2-a^2}} = \cosh^{-1}\left(\frac{x}{a}\right) + C, \qquad x > a > 0$

3. $\int \frac{dx}{a^2-x^2} = \begin{cases} \frac{1}{a}\tanh^{-1}\left(\frac{x}{a}\right) + C, & x^2 < a^2 \\ \frac{1}{a}\coth^{-1}\left(\frac{x}{a}\right) + C, & x^2 > a^2 \end{cases}$

4. $\int \frac{dx}{x\sqrt{a^2-x^2}} = -\frac{1}{a}\operatorname{sech}^{-1}\left(\frac{x}{a}\right) + C, \qquad 0 < x < a$

5. $\int \frac{dx}{x\sqrt{a^2+x^2}} = -\frac{1}{a}\operatorname{csch}^{-1}\left|\frac{x}{a}\right| + C, \qquad x \neq 0 \text{ and } a > 0$

$\frac{d}{dx}\left(\sinh^{-1}\left(\frac{x}{a}\right)\right) = \frac{1}{\sqrt{1+\left(\frac{x}{a}\right)^2}} \cdot \frac{d}{dx}\left(\frac{x}{a}\right)$

$= \frac{1}{a\sqrt{1+\left(\frac{x}{a}\right)^2}}$

$= \frac{1}{\sqrt{a^2\left(1+\left(\frac{x}{a}\right)^2\right)}}$

$= \frac{1}{\sqrt{a^2+x^2}}$

So, $\int \frac{1}{\sqrt{a^2+x^2}}\,dx = \sinh^{-1}\left(\frac{x}{a}\right) + C.$

Integration Formulas

Example: Evaluate

1. $\int \frac{2\,dx}{\sqrt{3+4x^2}}$

(1) Let $a^2 = 3 \Rightarrow a = \sqrt{3}$

$u^2 = 4x^2 \Rightarrow u = 2x \Rightarrow du = 2dx$

$\int \frac{2\,dx}{\sqrt{3+4x^2}} = \int \frac{du}{\sqrt{a^2+u^2}}$

2. $\int \frac{dx}{\sqrt{x^2+4}} = \sinh^{-1}\left(\frac{x}{2}\right) + C = \sinh^{-1}\left(\frac{u}{a}\right) + C$

$a^2 = 4 \Rightarrow a = 2$.

$= \sinh^{-1}\left(\frac{2x}{\sqrt{3}}\right) + C$.

3. $\int \frac{dx}{\sqrt{9x^2+4}}$ (u^2, a^2)

(3) Let $a^2 = 4 \Rightarrow a = 2$

$u^2 = 9x^2 \Rightarrow u = 3x \Rightarrow du = 3dx$

$\frac{1}{3}\int \frac{3dx}{\sqrt{9x^2+4}} = \frac{1}{3}\int \frac{du}{\sqrt{u^2+a^2}} = \frac{1}{3}\sinh^{-1}\left(\frac{u}{a}\right) + C$

$= \frac{1}{3}\sinh^{-1}\left(\frac{3x}{2}\right) + C$

4. $\int_0^{\frac{1}{2}} \frac{dx}{1-x^2}$ ($x^2 < 1$, $a^2 = 1$)

④ $\int_0^{1/2} \frac{dx}{1-x^2} = \left[\tanh^{-1} x\right]_0^{1/2} = \tanh^{-1}\left(\frac{1}{2}\right) - \tanh^{-1}(0)$

$= \tanh^{-1}\left(\frac{1}{2}\right).$